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Genesis of the newly discovered Pb–Zn vein of the Sucha giant fluorite deposit in Inner Mongolia: Constraints from LA-ICP-MS trace element of pyrite and sphalerite

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ABSTRACT

The Sumochaganaobao (Sucha) deposit in Siziwang Banner, Inner Mongolia is considered to be the largest single fluorite deposit known in the world, and apart from a small amount of localized pyritization, no metal mineralization has been found for many years. In recent years larger scale Pb–Zn polymetallic veins have been uncovered which indicates the potential for large scale metal mineralization in this deposit in addition to fluorite ore. Laser-ablation inductively coupled mass spectroscopy (LA-ICPMS) technique is used to investigate the distribution and substitution of trace elements in sphalerite and pyrite. The confirming results show that sphalerite is mainly rich in Fe, Mn, Cd, Co, Ni, Cu, In and poor in Sb and Mo. Pyrite is mainly rich in Ni, As, Sb, Cr, Mn, Co, Cu, Sn, Pb, Tl, Se and poor in Cd, Au, Ga, In, Bi. Iron, Mn, Cd, In, Co, Ni, Cr, Bi and Cu in sphalerite and Ni, As, Sb, Mn, Co, Cu, Ni, As, Pb, Tl, and Se in pyrite are generally lattice-bounded, while Pb and Cd present both in solid solution and/or discrete micro/nano-inclusions. The elemental contents of Fe, Mn and In and elemental ratios of Ga/In, Zn/Cd and In/Ge illustrate the trace elemental characteristics of medium temperature sphalerite. The trace elements in pyrite are diverse along with both relatively high-temperature elements (e.g. Ni, Co, Sn, Cu, etc.) and relatively low-temperature elements (e.g. As, Sb, Pb, etc.). The Co/Ni ratios indicate that the pyrite formed under a low-temperature condition. The crystallisation from sphalerite to pyrite may have undergone a relatively rapid cooling process from medium to low temperature. The characteristics of sphalerite and trace elements indicate that the genesis of this deposit is closest to that of SEDEX type deposit. Considering the marine sedimentation or marine volcanic eruption, jasperite development, clastic rock hosting, layered ore-body, and the geochemical characteristics of trace elements of this deposit, the newly discovered Pb–Zn vein should belong to the SEDEX type.

1. Introduction

As common minerals in Pb–Zn deposits, sphalerite and pyrite contain a wide variety of trace elements. For example, sphalerite often contains Fe, Mn, Cd, Ga, Ge, In, Se, Te and pyrite is rich in Au, Ag, Cu, Pb, Zn, Co, Ni, As, Sb, Hg, Bi, Se, Te, Tl. The type, content and occurrence state of the above elements contain a wealth of mineralization information (Liu et al., 1984; Large et al., 2009; Basori et al., 2018), which can be used not only to reconstruct hydrothermal evolution process (Genna and Gaboury, 2015), but also to indicate fluid composition and ore deposit genesis (Winderbaum et al., 2012; Reich et al., 2013; Basori et al., 2018). The trace elements as well as istpoes have been used to Characteriz geological processes, tracing material sources, and classify the genetic type of mineral deposits, to finely characterize the ore-forming processes and to evaluate the mineralization potentiality (Ye et al., 2012; Gong et al., 2011, 2016, 2022; Wu et al., 2021, 2022; Qi et al., 2022; zhang et al., 2022). Compared to the traditional single mineral selection test, LA-ICP-MS techniques has the characteristics of in-situ testing and high precision, which can obtain many information such as the true trace element data of sulfide and the occurrence state of trace elements, and it

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is the most effective method to study the trace element composition of sulfide (Qi et al., 2022).

Fluorite deposits can be developed in various tectonic settings, a wide range of geological environment and different deposit types (Han et al., 2020). It can be mineralized alone, or commonly associated with other metallic minerals. There are more than 40 fluorite deposits associated with lead, zinc, tungsten, tin, iron and rare earth in China (Wang et al., 2015). For example, the associated fluorite ore in Bayan Obo deposit is huge in scale, Shizhuyuan tungsten-tin-molybdenum-bismuth polymetallic deposit (Han et al., 2020), Taolin lead-zinc deposit (Yan, 1957), Huangshaping lead-zinc-tungsten deposit (Liu, 2007), etc. are all associated with sizeable fluorite resources. The Sucha deposit is considered to be the largest single fluorite deposit known in the world to date (Nie et al., 2008), the fluorite ore reserves are 2000×10^4 t. Because of the superior ore-forming conditions, the China-Mongolia border where the deposit is located and the neighboring areas have long been regarded as a target area for prospecting and exploration of polymetallic deposits (Wang et al., 2004), In the adjacent area of Mongolia in this region, large deposits such as Oyu Tolgoi super-large copper-gold deposit and Tsagaan Suvarga large copper-molybdenum deposit are developed (Li et al., 2016). It is worth mentioning that large deposits such as Oyu Tolgoi copper-gold deposit, Haramotu tin (tungsten) deposit and Har Tolgoi silver polymetallic deposit in southern Mongolia all have some degree of fluorite mineralization (Nie et al., 2010). Although

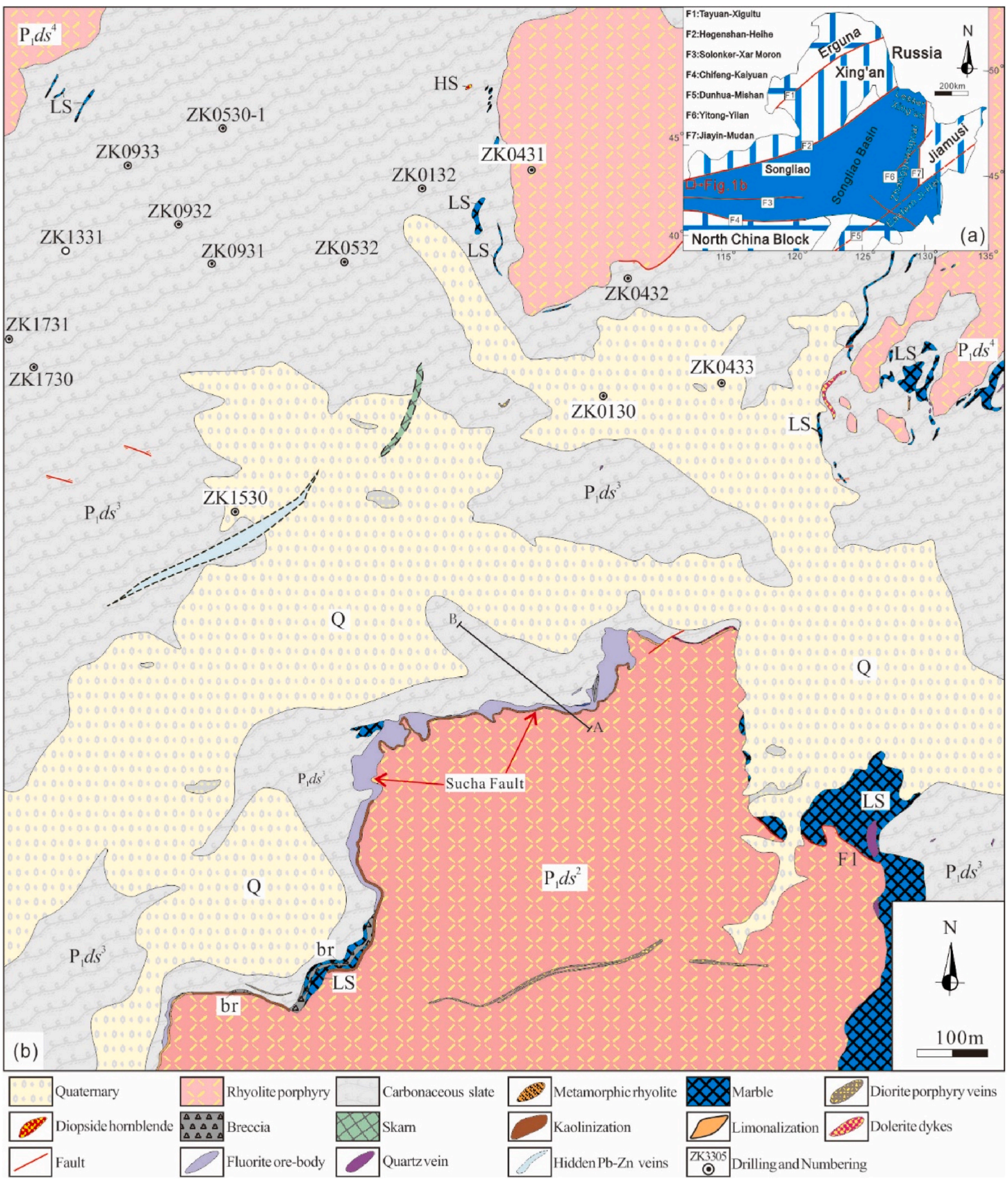

Fig. 1. Geological map of the Sucha deposit and its adjacent areas, Inner Mongolia
a- The structural diagram of Northeast China and its adjacent areas (modified after Zhang et al., 2010); b- Geological map of the Sucha deposit (modified after Pu et al., 2022).

in one side area of our country mainly develops fluorite deposits, the newly discovered metal ore deposits in this region, including Xilimiao manganese ore (Wang et al., 2011) and Dongjingcun copper polymetallic ore (Hou and Huang, 2016), indicate the metal mineralization potential in the area. The Sucha deposit has more than 40 years of mining history, in the early years, it was open-pit mining, but no large-scale metal mineralization, with the conversion to pit mining in later period, a large scale polymetallic vein of lead and zinc was revealed for the first time (Pu et al., 2022). This new discovery indicates that metal mineralization such as Pb–Zn is also present in this deposit except the fluorite ore, and the mineralization process is more complex. Although there have been many studies on the fluorite mineralization of the Sucha deposit, there is still a gap in the study of the newly discovered Pb–Zn polymetallic veins in this deposit.

In this paper, the study of mineralogy was completed firstly. Then the content of trace elements in Pyrite and Sphalerite was tested by LA-ICP-MS analysis technology. Thirdly, the composition and occurrence status of trace elements were identified. Finally, the mineralization temperature was estimated and the ore genesis of lead-zinc vein were constrained.

2. Geological setting

2.1. Regional geology

The Sucha super-large fluorite deposit is located in the northern of Siziwang Banner, Inner Mongolia (Fig. 1), at the border of China and Mongolia, geotectonically located in the eastern of the Central Asian Orogenic Belt (CAOB) (Xing'an-Mongolian orogenic belt), sandwiched between the Late Permian Solun suture of the Siberian (Xiao et al., 2003) and North China plates and the Erenhot-Hegenshan Deep Fracture (Fig. 1), and it belongs to the Inner Mongolia Surinher-Bainaimiao-Xilinhot Fe–Cu–Al–Pb–Zn metallogenic belt (Xu et al., 2008). The strata exposed in the region mainly include the Precambrian Elliger Temple Group metasedimentary rocks, the Lower Permian Dashizhai Formation volcanic-sedimentary rocks, the Upper Jurassic Chagauoer Formation volcanic rocks, the Upper Cretaceous sedimentary rocks and the Quaternary (Xu, 2009). Among them, the Lower Permian Dashizhai Formation is a set of marine acidic-intermediate-acidic volcanic-sedimentary clastic rocks and carbonates, distributed in north-east direction, with a total thickness of more than 9.4 km, which is the main ore-host strata in the region. The region developed various geological ages of Intermediate-acid intrusive rocks, they are mainly intruded into the volcanic-sedimentary rocks of the Dashizhai Formation and the volcanic rocks of the Upper Jurassic Chagauoer Formation in the form of batholith, stock, and rock groups. The superposition of the above-mentioned tectonic movements and multiple phases of magmatic mineralization provides a favorable mineralization space and plentiful sources for the regional mineralization including the Sucha deposit (Nie et al., 2008).

2.2. Deposit geology

The strata exposed in the Sucha deposit is mainly the Lower Permian Dashizhai Formation (P_1ds) and the Quaternary sediments (Fig. 1). The Dashizhai Formation is the main ore-host rock of the deposit (Fig. 2), mainly composed of rhyolite porphyry, tuff, tuff lava, carbonaceous-argillaceous slate, marble and limestone. The intrusive rocks of Late Garidonian, Hercynian and Yanshan periods are developed in the area, at the same time, quartz veins, diorite porphyrite veins and dolerite veins are also widely developed (Xu, 2009). There are striped jasperite developed in the southeast of the mining area (Fig. 3a and b). The Weijing granite (zircon U–Pb age 138 ± 4 Ma, Nie et al., 2009) exposes in the northwest of the deposit, distributed in a north-northeast direction and mainly consists of medium-fine grained granite porphyritic granite, moyite and fine-grained granite. Sucha compressive-torsional fault is the main ore-controlling structure of the Sucha deposit, which is the interlayer fault zone of the second lithological segment (rhyolite

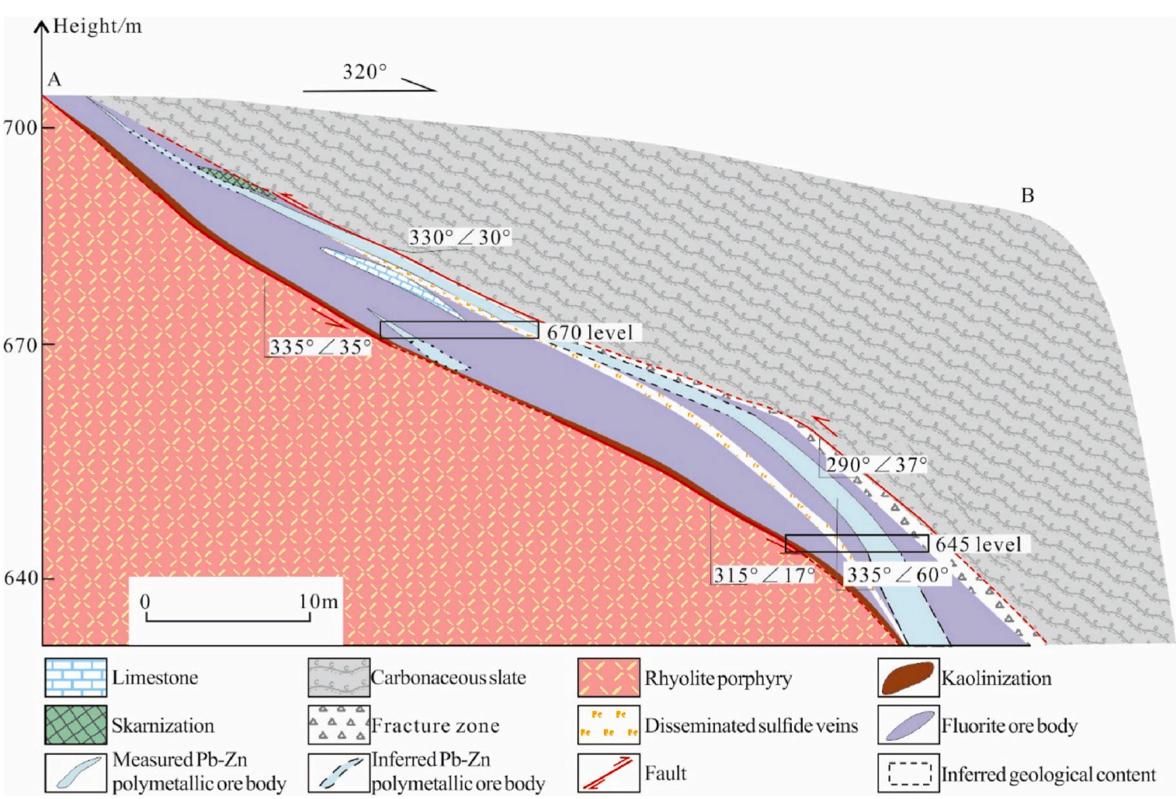

Fig. 2. Section of Pb–Zn polymetallic veins and fluorite ore bodies in the Sucha deposit.

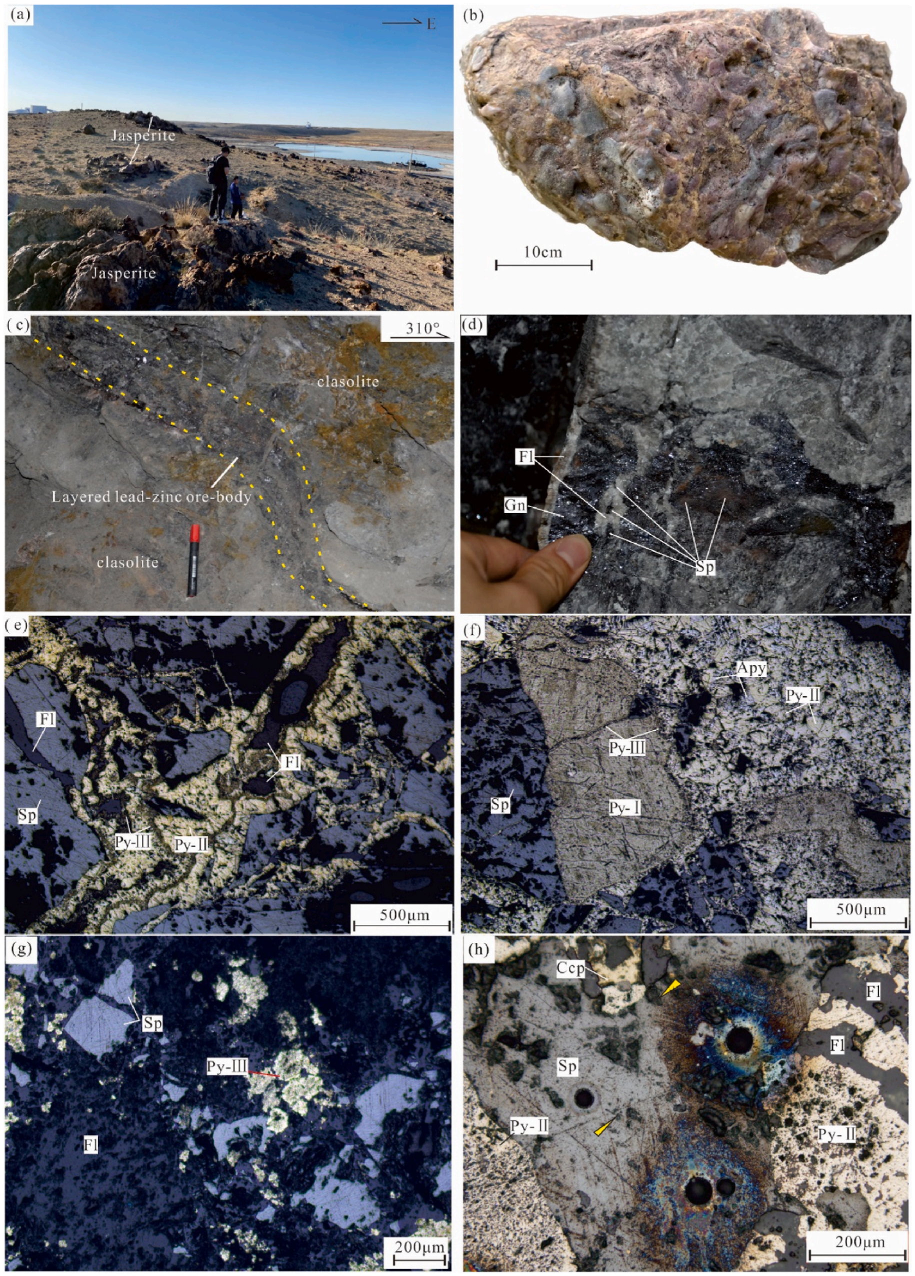

Fig. 3. Rock, ore-body and main minerals at the Sucha deposit
a- Striped jasperite outcrop; b- Jasperite; c- Layered Pb–Zn vein at 670 level; d- The lead - zinc ore are structural brecciform encased, interspersed and cemented by white fluorite; e– Sphalerite is cataclastic and cemented by pyrite veins, and pyritization occurs in multiple stages, all of which are veins; f- Pyrite of the first generation has a good euhedral structure and arsenopyrite develops in pyrite of the second generation as a skeleton crystal; g- Cataclastic sphalerite cemented by fluorite; h-multiperiod pyrite is developed.

porphyry) and the third lithological member (carbonaceous-argillaceous slates and lenticular limestone) of Dashizhai Formation. The scale and geometry of the fluorite ore in the Sucha deposit is entirely controlled by the Sucha Fault, with an overall North-East distribution, locally shows anti-“S” and “V”-shape, the thickness of the ore body varies from several meters to 40 m with the width of the fault zone (Nie et al., 2008). The main near-ore wall rocks alteration is kaolinization, silicification, chloritization, pyritization, skarnization and marbleization. The newly discovered Pb–Zn veins in the 645 level and 670 level are developed in clastic rock and fluorite ore body controlled by the Sucha Fault Zone and are consistent with the occurrence of fluorite ore body, trending NE and NW, the dip angle varies between 30° and 40°. It is inferred that the three middle sections Pb–Zn veins are likely to be continuously developed based on their occurrence. The Pb–Zn ore bodies are Layered, vein-like and lenticular (Figs. 2 and 3c and d), with good continuity and localized fragmentation, and are approximately 0.5 m–2 m wide.

3. Sampling and analytical methods

3.1. Samples

The Pb–Zn ore samples in this study were all collected from the 645 level and 670 level. Except for the 645 level and 670 level, the pyrite bearing samples mostly came from drill cores. The Pb–Zn ore structure is predominantly disseminated, brecciated and reticulated, and the mineral structure is mainly authomorphic-hypidiomorphic granular, skeletal crystalline and fine veined. The ore minerals are mainly sphalerite, galena, pyrite, chalcopyrite and poisonous sand, with small amounts of bismuthite, cassiterite and freibergite (Fig. 3e、f、g、h). Mineralogical characteristics show that the pyrite in the Sucha deposit has three generations, respectively, authomorphic or hypidiomorphic pyrite (PyI), vein pyrite (PyII) and fine veined, rimmed structure pyrite (PyIII), while sphalerite shows the characteristics of single stage of mineralization (Fig. 3), in the form of allotriomorphic texture or metasomatic relict texture, often punctured or metasomatic by vein pyrite and fluorite. The formation sequence of the main minerals in the deposit is galena → sphalerite + authomorphic or hypidiomorphic pyrite (PyI) → cassiterite, bismuthite → vein pyrite (PyII) → freibergite, chalcopyrite → arsenopyrite → fine veined rimmed structured pyrite (PyIII) → off-white fine-grained fluorite (Pu et al., 2022).

3.2. Analytical methods

The analysis of trace elements in sulfide was completed at the National Testing Centre of the Chinese Academy of Geological Sciences. Using a Thermo Element II single-receiver high-resolution magnetic sector ICP-MS coupled to a New Wave UP 213. The exfoliated material was carried by Helium. The working conditions of the instrument are 40 μm spot size with a repetition rate of 10 Hz, 10–30 J/cm^2 laser flow and energy of 0.176 mJ. During the analysis process, first 15 s of background value acquisition, then 45 s of sample denudation, and finally 15 s of cleaning the system (Fig. 4), Specific analysis process and technology refer to Yuan et al. (2015). Analysis elements include Li, Be, B, Na, Mg, Al, Si, P, S, K, Ca, Sc, Ti, V, Cr, Mn, Co, Ni, Cu, Ga, Ge, As, Se, Rb, Sr, Y, Zr, Nb, Mo, Ag, Cd, In, Sn, Sb, Te, Cs, Ba, La, Ce, Pr, Nd, Sm, Eu, Gd, Tb, Dy, Ho, Er, Tm, Yb, Lu, Hf, Ta, W, Re, Au, Tl, Bi, Pb, Th, U. The analytical calculations were calibrated using the international standard Mass-1 (Wilson et al., 2002), the analysis precision was 0.1×10^{-6}.

4. Results

The main trace elements that were riched are Cr, Mn, Co, Ni, Cu, Zn, As, Mo, Ag, Sn, Sb, Au, Bi, Pb and Rare-dispersed elements Ga, Ge, Cd, In, Tl, Se, while the Te content below detection limits. LA-ICP-MS tests carried out on a total of 32 points of various types of pyrite in Pb–Zn polymetallic ores from the Sucha deposit, showed the following characteristics for trace element content (Fig. 5a)：(1)Rich in Ni, As and Sb. Ni is mainly enriched in Py II pyrite and Py III pyrite, and Py II pyritehaves a higher Ni content than Py III pyrite, both of them are much higher than Py I pyrite, indicating that the Ni content underwent a dramatic increase followed by a slight decrease as mineralization progressed. The As content varied from PyI→ PyII→ PyIIIshowing a significant increasing trend; the Sb content did not change much, but also showed a increasing trend from PyI→ PyII→ PyIII; (2)Less rich in Cr, Mn, Co, Cu, Sn, Pb, Tl and Se. The content of Cr increased first and then decreased from PyIto PyIIto PyIII; Mn content varied from PyIto PyIIto PyIII, showing an obvious decreasing trend; the contents of Co, Cu and Sn increased first and then decreased with the progress of mineralization. Pb is mainly riched in Py I and Py II, and the Pb content in Py III pyrite is sharply reduced; Se has a similar concentration regularity to Pb, mainly enriched in PyIand PyII, and PyIII has a lower Se content; Tl shows the opposite concentration regularity to Pb and Se, mainly enriched in Py II and Py III, while Tl content in PyIis very low; (3) Contains small amounts of Zn, Ge, Ag and Mo. The contents of Zn, Ge and Ag all go through a process of increasing and then decreasing. Mo content shows decreasing characteristics as mineralization process, mainly enriched in PyI; (4) Poor in Cd, Au, Ga, In and Bi. The content of Cd and Bi first increase and then decrease; Au content is slightly higher in PyIIGa content decreases; In content increases.

A total of 25 points of sphalerite were tested by LA-ICP-MS. In general, the trace element composition of sphalerite from the Sucha deposit has the following characteristics as showing in Table 1 and Fig. 5b: (1) Rich in Fe, and with a wide range of variation. (2)Rich in Mn and Cd,

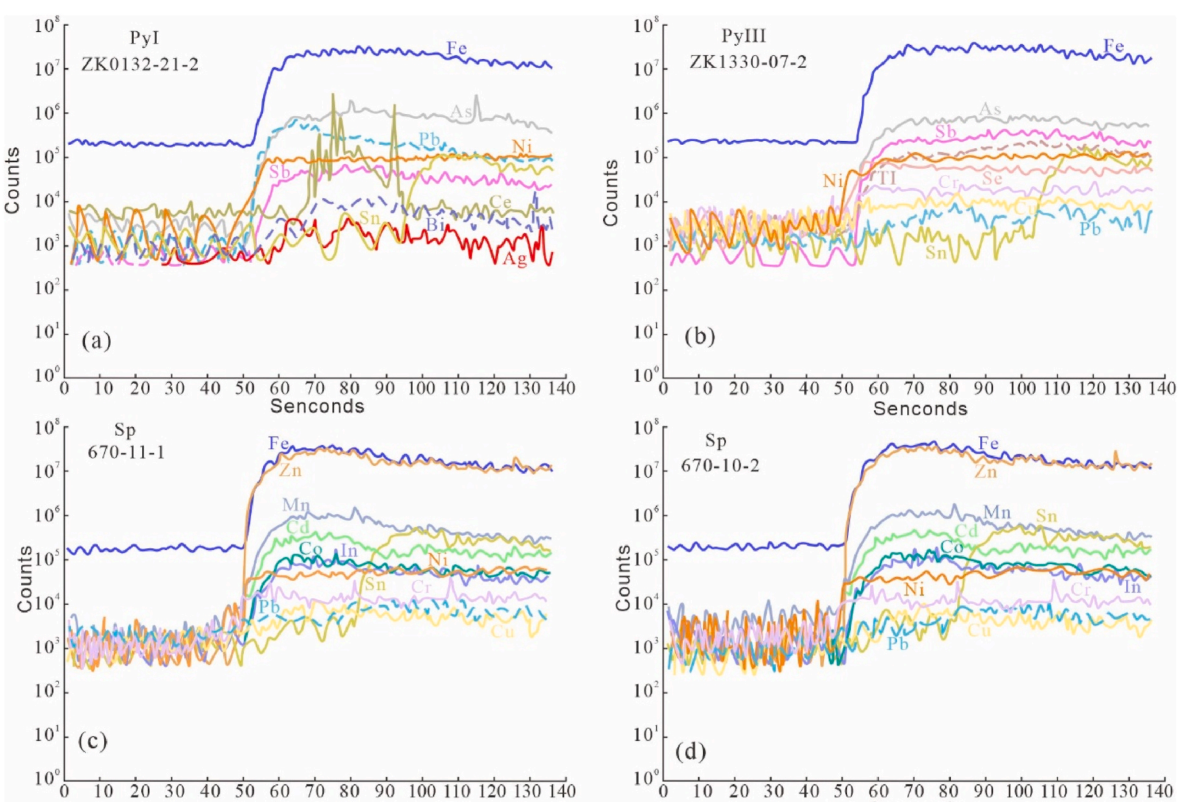

Fig. 4. Representative time resolved LA-ICP-MS depth profiles for Pyrite and sphalerite.

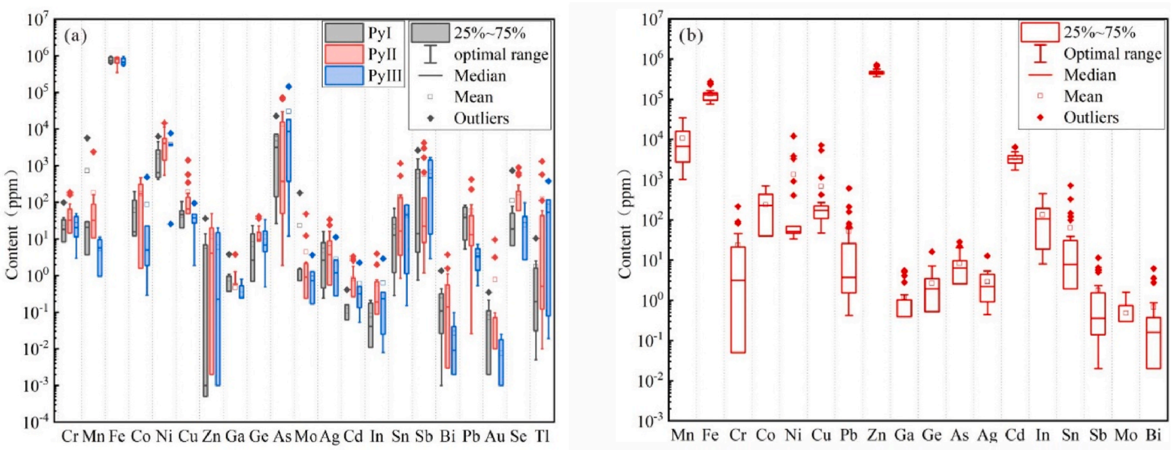

Fig. 5. Box-and-whisker plots for different generations of pyrites (a) and sphalerite (b) in-situ trace elements from the Sucha deposit.

Table 1
Average concentration of elements for pyrite and sphalerite of the Sucha deposit (ppm).

PyI											
Element	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Mo
Avg.	27.41	727.55	/	59.70	1791.83	44.41	6.25	0.81	7.16	5440.07	23.21
Element	**Ag**	**Cd**	**In**	**Sn**	**Sb**	**Bi**	**Pb**	**Au**	**Se**	**Tl**	
Avg.	4.80	0.11	0.08	21.85	531.65	0.28	41.62	0.07	111.08	1.76	
PyII											
Element	**Cr**	**Mn**	**Fe**	**Co**	**Ni**	**Cu**	**Zn**	**Ga**	**Ge**	**As**	**Mo**
Avg.	58.15	184.99	/	174.26	4413.98	192.25	11.65	0.47	13.72	13242.64	4.42
Element	**Ag**	**Cd**	**In**	**Sn**	**Sb**	**Bi**	**Pb**	**Au**	**Se**	**Tl**	
Avg.	7.42	0.81	0.59	140.94	584.50	0.51	57.95	0.77	179.80	125.34	
PyIII											
Element	**Cr**	**Mn**	**Fe**	**Co**	**Ni**	**Cu**	**Zn**	**Ga**	**Ge**	**As**	**Mo**
Avg.	24.14	5.50	/	87.54	3760.94	41.11	5.91	0.30	11.59	30039.73	1.09
Element	**Ag**	**Cd**	**In**	**Sn**	**Sb**	**Bi**	**Pb**	**Au**	**Se**	**Tl**	
Avg.	2.75	0.60	0.63	43.77	673.76	0.03	3.55	0.01	24.10	100.71	
Sp											
Element	**Mn**	**Fe**	**Cr**	**Co**	**Ni**	**Cu**	**Pb**	**Zn**	**Ga**		
Avg.	10,715.87	132,781.06	23.61	235.37	1349.14	682.81	52.15	/	1.20		
Element	**Ge**	**As**	**Ag**	**Cd**	**In**	**Sn**	**Sb**	**Mo**	**Bi**		
Avg.	2.68	8.26	2.91	3486.96	133.93	64.29	1.75	0.48	0.67		

with a narrow range of variation. (3)Relative rich in Co, Ni, Cu, In, with a narrow range of variation. (4) Containing small amounts of Cr, Sn and Pb, with a wide range of variation. (5) Containing small amounts of Ag, Bi, Ga, Ge and As, with a narrow range of variation. (6) Poor in Sb and Mo.

5. Discussions

5.1. Trace element distribution in pyrite

Siderophile and chalcophile elements, such as Co, Ni, Se, Te and As enter the lattice of pyrite via isomorphism (Tribovillard et al., 2006; Zhang et al., 2014a, 2014b). Nickel and Co can enter the pyrite lattice via isomorphous replacement of Fe, whereas As, Se and Te enter the lattice of pyrite by replacing S (Large et al., 2009; Koglin et al., 2010). The trace element species in pyrite are similar to those in sphalerite, but the general contents of Ni, As, Sb,Cr, Sn, Tl, Se and Pb are higher than those in sphalerite, while Cd and In are lower. The signals of Ni, As, Sb, Cu, Pb, Tl, and Se are flat (Fig. 4a and b), indicating that these elements mainly enter the pyrite lattice as solid solutions, while the signals of Sn and Ce have large fluctuations (Fig. 4a and b), indicating that they mainly enter the pyrite crystal as microinclusions.

Cobalt and Ni can substitute Fe by a coupled mechanism ($2Fe^{2+} \leftrightarrow Co^{2+} + Ni^{2+}$) and have a strong correlation (Fig. 6a), However, considering that all samples have Co/Ni ratios much less than 1, it is more convincing that Co and Ni are directly substituted by Fe ($Fe^{2+} \leftrightarrow Co^{2+}, Ni^{2+}$). The strong correlation between Co and Ni may be attributed to temperature, sulfur fugacity, original fluid composition and other factors (Maslennikov et al., 2009). Se and As have a negative correlation (Fig. 6b), indicating that they inhibit each other from entering the pyrite lattice, which suggests that As and Se enter the lattice of pyrite mainly by directly replacing S. Sb, As and Tl + Cu + Ag have a good correlation (Fig. 6c and d), suggesting a possible coupled substitution mechanism of $(Tl^{+} + Cu^{+} + Ag^{+}) + (Sb^{3+}, As^{3+}) \leftrightarrow 2Fe^{2+}$ (Yang et al., 2022). However, it is also possible that As substitutes S during the deformation of pyrite (Deditius et al., 2014; Kesler et al., 2010).

The much larger ionic radius of Pb^{2+} compared to Fe^{2+} makes it hard to enter the pyrite lattice. (Yang et al., 2022; Ye et al., 2011). However, the presence of As in pyrite can distort the pyrite lattices and allow Pb to get into these lattices and substitute with Fe (Deditius et al., 2014; Gregory et al., 2015). The smooth Pb signal in some samples (Fig. 4a), as well as the certain correlation between Pb and Sb (Fig. 6e) indicate that small amounts of Pb can be incorporated into the pyrite structure via a direct substitution of $Fe^{2+} \leftrightarrow Pb^{2+}$ and other coupled substitution mechanisms. However, most of the Pb should occur in pyrite as micro-inclusions, supported by some ragged signals curves of Pb and the observation of galena micro-inclusions.Sn can enter the lattice or occurs as oxide inclusion in pyrite, most of the Sn should occur in pyrite as

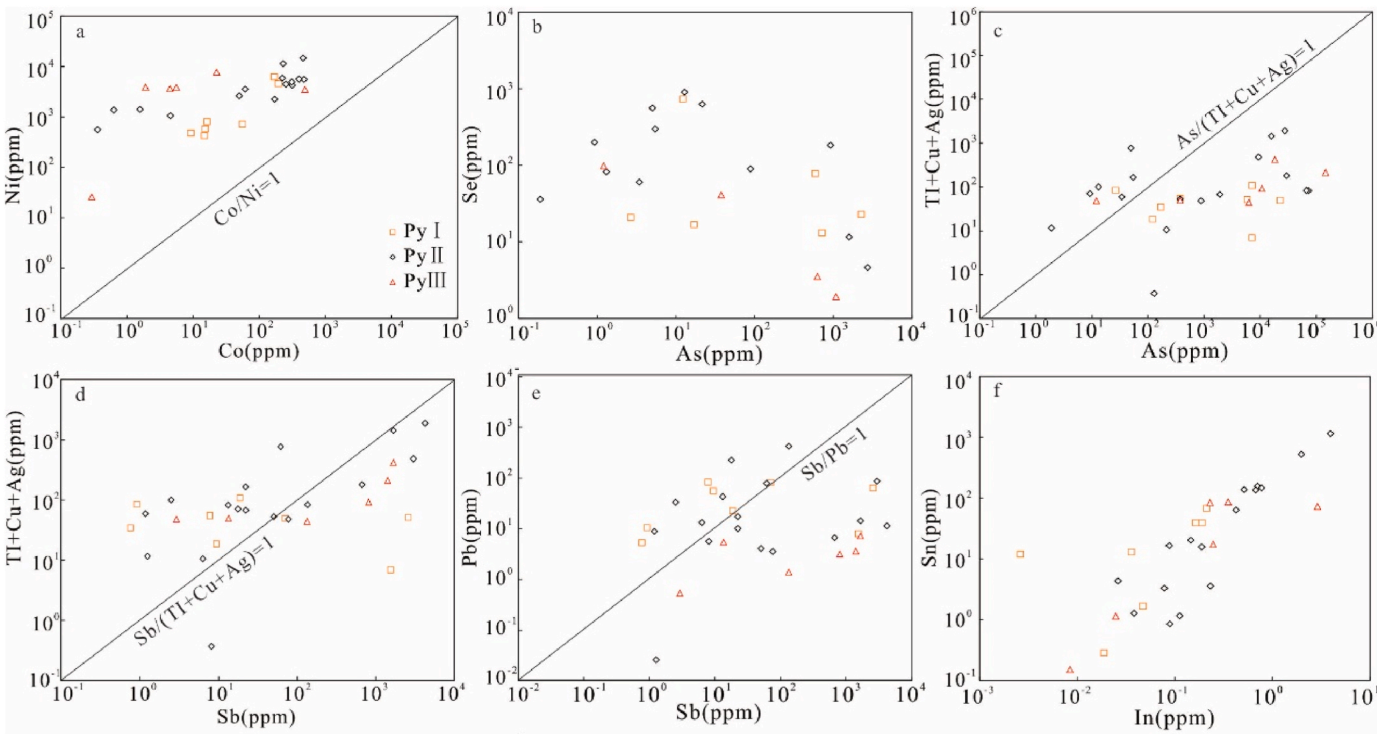

Fig. 6. Trace elements covariation diagram of pyrite in the Sucha deposit.

micro-inclusions, supported by ragged signals curves of Sn (Fig. 4a and b) and the observation of cassiterite micro-inclusions (Pu et al., 2022). However, positive correlation between Sn and In (Fig. 6f), indicating coupled substitution mechanisms and similar occurrence of Sn and In.

5.2. Trace element distribution in sphalerite

The substitution of Zn^{2+} by trace elements in sphalerite predominantly occurs by simple and coupled substitution mechanisms. Divalent cations with ionic radius similar to Zn^{2+}, such as Fe^{2+}, Cu^{2+}, Mn^{2+}, Co^{2+}, Ni^{2+}, Cd^{2+}, Ge^{2+}, can be directly substituted into the sphalerite crystal lattice ($M^{2+} \leftrightarrow Zn^{2+}$). while monovalent, trivalent and tetravalent cations, such as Ag^{+}, Cu^{+}, In^{3+}, Ga^{3+}, As^{3+}, Sb^{3+} Sn^{+} and Ge^{+}, have large ionic radius and different valences, generally get into the sphalerite lattice through coupled mechanisms (Cook et al., 2009a; Lockington et al., 2014), such as $Cu^{+} + Sb^{3+} \leftrightarrow 2Zn^{2+}$ and $Ag^{+}+Cu^{+}+Sn^{4+} \leftrightarrow 3Zn^{2+}$. In addition, some trace elements occur in sphalerite as micro-inclusions. Generally, the element signals in the time-resolved depth profles can be used to infer how the element occur in sphalerite (Cook et al., 2009a). Excluding the uniform distribution of mineral micro-inclusions at the ablation scale, smooth signal profles generally indicates that the element is lattice-bound (solid solution), while ragged profiles and high variability between individual ablation spots indicate the presence of micro-inclusions and/or nanoparticles.

The element signals in the time-resolved depth profles of the Sucha show that Mn, Fe, In, Co, Ni, Cr and Cu have smooth signal profles (Fig. 4c and d), indicating that these elements enter the sphalerite lattice mainly as solid solutions, while Sn has more fluctuating signal profles (Fig. 4c and d), indicating that Sn enter as micro-inclusions into the sphalerite, which is consistent with the fact that cassiterite was discovered in this deposit (Pu et al., 2022). Pb and Cd have both smooth signal profles and fluctuating signal profles (Fig. 4c and d), indicating that Pb and Cd enter the sphalerite lattice as solid solutions and micro-inclusions into the sphalerite.

Numerous studies have shown that Mn^{2+}, Fe^{2+}, Co^{2+}, Ni^{2+} and Cd^{2+} substituted for Zn^{2+} in the sphalerite lattice through simple and direct substitution mechanisms (Torró et al., 2022; Wei et al., 2021a–c; Yuan et al., 2018). The smooth Mn, Fe, and Cd signal curves of sphalerite indicate that Mn^{2+}, Fe^{2+}, and Cd^{2+} directly enter the sphalerite lattice and substitute with Zn^{2+} (Fe^{2+}, Mn^{2+}, $Cd^{2+} \leftrightarrow Zn^{2+}$). In Fig. 7a, most Fe and Mn at the Sucha deposit show a signifcant positive correlation, indicating that they promote each other to enter the sphalerite lattice, which suggests the coupled mechanism $2Zn^{2+} \leftrightarrow Fe^{2+} + Mn^{2+}$(Yang et al., 2022). In Fig. 7b, Fe and Cd also have some negative correlation, indicating that they also have some inhibitory relationship. Co + Ni and Fe have no obvious linear relationship (Fig. 7c), suggesting that Co and Ni enters the sphalerite lattice through substitution mechanism of $Zn^{2+} \leftrightarrow Co^{2+}$, Ni^{2+} rather than $Fe^{2+} \leftrightarrow Co^{2+}$, Ni^{2+}.

Cu usually enters the sphalerite lattice through being directly substituted into the sphalerite crystal lattice ($Cu^{2+} \leftrightarrow Zn^{2+}$)(Liu et al., 1984), as well as a coupled mechanisms of $Cu^{+} \leftrightarrow Sb^{3+}+2Zn^{2+}$, $2(Ag, Cu)^{+}+Sn^{+} \leftrightarrow 3Zn^{2+}$, $2Zn^{2+} \leftrightarrow Cu^{+} + Ga^{3+}$ and $3Zn^{2+} \leftrightarrow 2Cu^{+} + Ge^{+}$ (Cook et al., 2009a). The correlation between Cu + Sn and Ag is significant (Fig. 7d), suggesting that Cu enter sphalerite through substitution mechanisms of $2(Ag, Cu)^{+}+Sn^{+} \leftrightarrow 3Zn^{2+}$. The data points above the line of Cu + Sn/Ag = 1 suggests that this equation is not adequate for the intake of Cu and could be explained by the substitution mechanism of $Cu^{2+} \leftrightarrow Zn^{2+}$. Indium, generally occurs as In^{3+}, enters into sphalerite throug coupled substitution (Belissont et al., 2014). Some authors proposed possible mechanisms, such as $3Zn^{2+} \leftrightarrow In^{3+} + Sn^{3+} + \square$ (where $\square$ denotes a vacancy, Belissont et al., 2014) and $3Zn^{2+} \leftrightarrow In^{3+} + Sn^{2+} + (Cu, Ag)^{+}$ or $4Zn^{2+} \leftrightarrow In^{3+} + Sn^{4+} + (Cu, Ag)^{+} + \square$ (Murakami and Ishihara, 2013). The mechanism of $2Zn^{2+} \leftrightarrow Cu^{+} + In^{3+}$ is also reported (Cook et al., 2009a), particularly in high In contents sphalerite if under reducing conditions in hydrothermal fluids. A positive correlation between Cu, In + Ga + Ge, In + Ga + Ge + Sn and Cu + Ag (Fig. 7e and f) reflect structural substitution of $2Zn^{2+} \leftrightarrow Cu^{+} + In^{3+}$ and other coupled substitution mechanisms with Ag^{+}, Cu^{+}, Ga^{3+}, Ge^{4+}, and Sn^{4+}(Bauer et al., 2019), while the substitution with $3Zn^{2+} \leftrightarrow In^{3+} + Sn^{3+} + \square$ could be negligible.

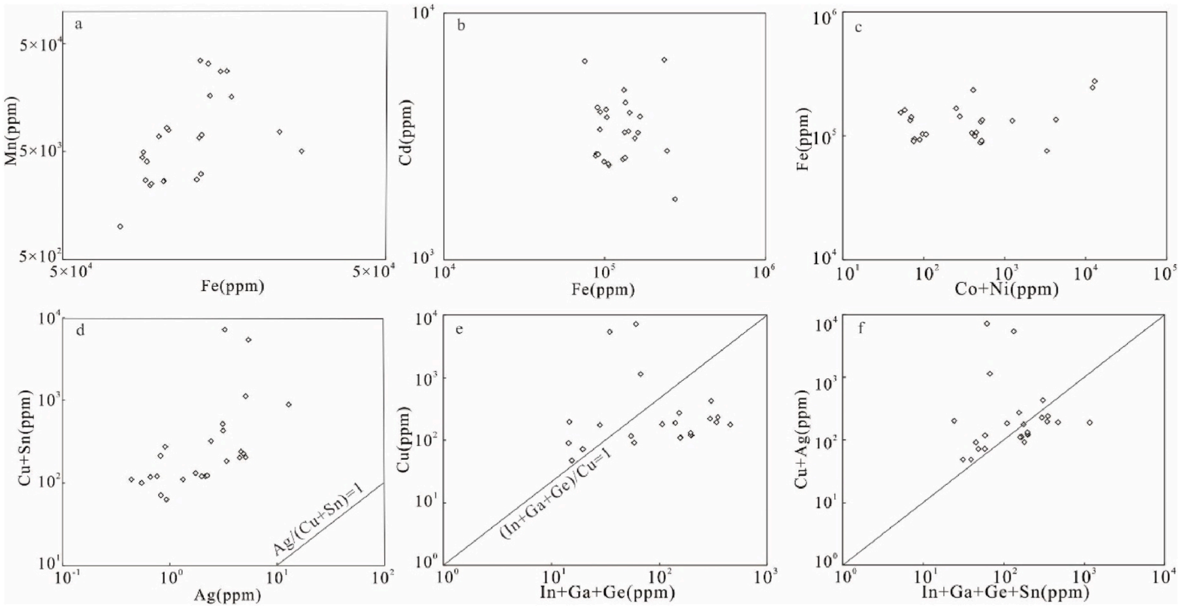

Fig. 7. Trace elements covariation diagram of sphalerite in the Sucha deposit.

5.3. Temperature of the Pb–Zn mineralization

Due to the different metallogenic temperature, the trace elements in sphalerite also show regular variations. Ionic activity is enhanced and isomorphism substitution of similar ionic radii can take place easily at high temperature, so sphalerite formed at high temperature is relatively enriched in Fe, Mn, In, Se and Te, and has high Zn/Cd values, while low-temperature-sphalerite is rich in Cd, Ga and Ge, and has low Ga/In and Ge/In values (Liu et al., 1984, 2008; Zou et al., 2012). As showing in Table 2, the Fe content of the sphalerite in this deposit is similar to the Dulong silica-type Pb–Zn deposit, and the Laochang Pb–Zn deposit of medium-temperature. The In content is well above that of the Mayuan Pb–Zn deposit of low temperature, close to the Laochang Pb–Zn deposit, Huaniushan Pb–Zn deposit, Sinongduo Pb–Zn deposit of medium temperature and below that of the Dulong Sn deposit of high temperature, indicating that the Sucha deposit formed in a medium temperature environment. The Ga/In, Ge/In and Zn/Cd values of sphalerite can also be used to determine the deposit formation temperature: when Ga/In $<$ 0.01 and Zn/Cd $>$ 500, it is a high temperature condition ($>$300 °C); when Ga/In values are 0.01–5.00 and Zn/Cd values are between 100 and 500, it is a medium temperature condition (200–300 °C); when Ga/In value is 1.0–100, Zn/Cd $<$ 100, it is low temperature condition ($<$200 °C) (Liu et al., 1984; Han, 1994; Zhu et al., 1995). The Ga/In ratio and the Zn/Cd ratio range of this deposit are consistent with the values of medium-temperature sphalerite.

The type and content of trace elements in pyrite is controlled by temperature, usually showing that the higher the ore-forming temperature, the more types and higher contents of trace elements. Some temperature-sensitive elements such as As, Zn, Sb, and Pb are mainly enriched in pyrite formed under low-temperature condition while Se, Sn, Te, Co, Bi, U, Mo, Cu and Ni are concentrated at mid to high temperature under highly reducing conditions (Keith et al., 2016; Liu et al., 2021). Moreover, the Co/Ni ratio of pyrite can also be used as a temperature indicator: when Co/Ni less than 1, the pyrite is formed under low-temperature conditions, and when Co/Ni $>$ 1 the pyrite is formed under high-temperature conditions (Cook, 1996; Maghfouri et al., 2018). The pyrite of the Sucha deposit is rich in Ni, As and Sb, less rich in Cr, Mn, Co, Cu, Sn, Pb, Tl and Se, with minor amounts of Zn, Ge, Ag and Mo, and poor in Cd, Au, Ga, In and Bi. It is evident that the pyrite of this deposit has a wide range of trace elements, and the enriched elements include both relatively high temperature elements (e.g. Se, Ni, Co, Sn Cu, etc.) and relatively low temperature elements (e.g. As, Sb, Pb, etc.). The Co/Ni ratios of pyrite formed in different mineralization stages are similar and much less than 1, suggesting that the pyrite in this deposite was formed under a low-temperature condition.

The types of trace elements in sphalerite and pyrite in the Sucha deposit are diverse and similar, the trace element characteristics of sphalerite indicates that it was formed under the medium temperature condition, but the trace element Co/Ni value of pyrite is much less than 1, indicating a relatively low temperature genesis. Both relatively high temperature and relatively low temperature elements are enriched in sphalerite and pyrite, suggests that the crystallisation from sphalerite to pyrite may have undergone a relatively rapid cooling process from medium to low temperature.

5.4. Implications for ore genesis and metallogenic

The trace element characteristics of pyrite and sphalerite can reflect the deposit genetic type.Sphalerite in the Sucha deposit is characterized by enrichment in Fe, Mn and In, and depletion in Ga, Ge and Sb, comparing with significant types of deposits in the world, it generally exhibits trace element characteristics similar to those of VMS-type and

Table 2
Average characteristic values of Sphalerite in the Sucha deposit and typical Pb–Zn deposits in China.

Deposit (Type)	Fe/10^{-2}	Mn/10^{-6}	In/10^{-6}	Cd/10^{-6}	In/Ga	In/Ge	Zn/Cd	Cd/Fe	Cd/Mn	Reference
Mayuan (MVT)	0.29	47.82	6.30	1002.45	0.07	0.008	\	0.35	20.96	Hu et al., 2014
Dulong (Skarn)	> 10	1600	426	1625	283.32	645.47	290.77	\	1.21	Ye et al. (2017)
Sinongduo (Hydrothermal)	1.02	733.40	209.18	3860.99	130.80	1135.81	0.01	0.38	5.26	Xiao et al., 2003
Huaniushan (SEDEX)	7.58	10100	269.77	4481.82	53.27	646.50	125.56	0.06	0.44	Kang et al. (2020)
Laochang (VMS)	12	2840.5	188.5	6452.5	13.64	340	133	0.05	2.27	Ye et al. (2012)
Sucha	13.28	10715	133.93	3487	214.45	111.73	145.89	0.03	0.84	This study

skarn type deposits, while differing markedly from stratabound deposits (Fig. 8). As showing in Table 2, the trace element characteristics show similarities to SEDEX, VMS and magmatic-hydrothermal type deposit, and significant differences from those of MVT-type deposit. The Cd content of Sphalerite in the Sucha deposit is closer to that of the SEDEX, VMS and magmatic-hydrothermal type deposits. The disperse elements Tl is mainly enriched in low-temperature sphalerite and can effectively distinguish between Stratabound type and magmatic-hydrothermal Pb–Zn deposits (Maslennikov et al., 2009). In general, the mass fraction of Tl in sphalerite of magmatic-hydrothermal deposits is $< 6 \times 10^{-6}$ (Xie, 1982), and the Tl content of sphalerite of the Sucha deposit ranges from 0 to 0.57×10^{-6}, it shows the characteristics of magmatic hydrothermal deposit.

The Cd/Fe, Cd/Mn and Zn/Cd ratios can also clarify whether the mineralization process is related to magmatic hydrothermal activity, with Cd/Fe, Cd/Mn and Zn/Cd ratios of sphalerite of Pb–Zn deposits related to magmatic activity being less than 0.1, 5 and 250, respectively, while Cd/Fe, Cd/Mn and Zn/Cd ratios of sphalerite of sedimentary or Stratabound type Pb–Zn deposits are greater than 1, 10 and 400, respectively (Cao et al., 2014; Tian et al., 2015). In addition, Hu et al. (2014) concluded that sphalerite in Pb–Zn deposits associated with magmatic or volcanic activity is characterized by high In and low Ge, while medium-low temperature sphalerite associated with basin brines is enriched in Ge and poor in In. The Cd/Fe, Cd/Mn and Zn/Cd ratios of sphalerite in the Sucha deposit and the characteristic of high In and low Ge show characteristics of magmatic-hydrothermal deposit. Previous studies have shown that Mn–Ag, Mn–In + Sn, Fe–Mn, Mn–In/Cd, In/Ge–Mn, In–Mn/Fe and Ga–In discrimination diagrams can effectively distinguish between different genetic types of sphalerite (Oyebamiji et al., 2022), and in each of these diagrams, the sphalerite cultellation of the Sucha deposit mainly fall within or near the SEDEX and VMS-type deposits (Fig. 9).

Pyrite in hydrothermal mineralization systems is often enriched in some important trace elements such as Cu, Pb, Zn, As, Ag, Au, Co, Ni, Sb, Se, Te, Tl and Bi(Cook et al., 2009b; Reich et al., 2013), which directly record the nature of the mineralising fluids. For example, sedimentary pyrite is usually rich in Zn and poor in Co and Ni(Mukherjee and Large, 2017; Li et al., 2019a), Pyrite formed in magmatic system is usually rich in Co and Se, and poor in Sb and As (Dare et al., 2011; Duran et al., 2015), The pyrite formed in the hydrothermal system is usually relatively rich in Sb and As and poor in Co and Se (Large et al., 2009; Revan et al., 2014; Patten et al., 2016). The pyrite of the Sucha deposit is rich in

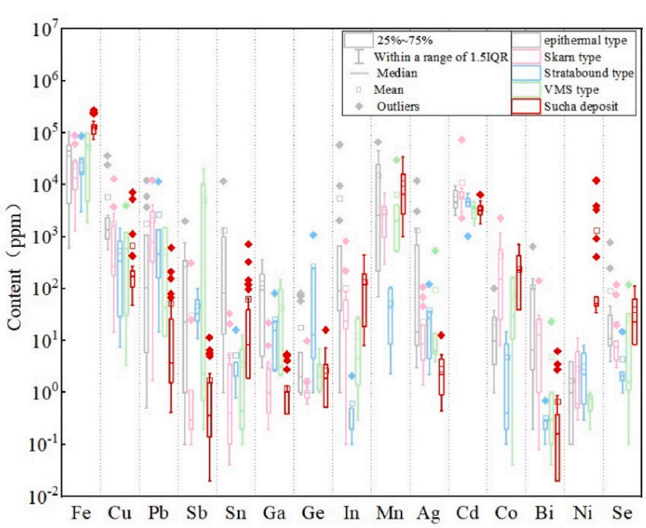

Fig. 8. Box-and-whisker plots for comparison between sphalerite trace elements in the Sucha deposit and other types of deposits
(Data on other types of deposits are from Cook et al., 2009a).

Ni, As and Sb, less enriched in Cr, Mn, Co, Cu, Sn, Pb, Tl and Se, with minor amounts of Zn, Ge, Ag and Mo, and poor in Cd, Au, Ga, In and Bi, showing trace element characteristics similar to those of pyrite formed in magmatic and hydrothermal systems. Co and Ni are often substituted for Fe in pyrite in the form of isomorphism, and the variation in Co and Ni content is mainly controlled by the physicochemical conditions during pyrite precipitation, thus different genetic types of pyrite usually have different Co/Ni ratios and are indicative of their formation environment (Bralia et al., 1979). Pyrite associated with magmatic hydrothermal fluids typically has a high Co/Ni ratio (Co/Ni $>$ 1), while sedimentogenic pyrite has a low Co/Ni ratio (Co/Ni $<$ 1) (Cook et al., 2009b). In the Co–Ni covariance diagram, all of the pyrite in the Sucha deposit is cast within or near the SEDEX-type deposit range (Fig. 10).

The lead-zinc deposits of different genetic types in China have certain regional distribution, such as MVT type and Huize type lead-zinc deposits are mostly developed in the Sichuan-Yunnan-Guizhou area (Wu et al., 2021; Zhao et al., 2022; Wang et al., 2023), while SEDEX type and VMS type lead-zinc deposits are mostly developed in the northwest and northeast of China (Zhang et al., 2014c). High-precision isotopic chronology and geochemical studies indicate that the North China Plate collided with the Siberian Plate during the late permian-Early Triassic and eventually led to the closure of the Paleo-Asian Ocean (Xiao et al., 2003), and the rhyolitic porphyry of floor surrounding rock at the Sucha deposit, with a zircon U–Pb age of 276 $\pm$ 10 Ma (Nie et al., 2009), is the product of medium-acidic magmatism in Early Permian Paleo-Asian interoceanic extensional tectonic setting. The Lower Permian Dashizhai Formation as surrounding rock is a set of marine acidic-medium acidic volcanic-sedimentary clastic and carbonate rocks. What is more, jasperite, representing the products of hydrothermal sedimentation, commonly present in SEDEX type deposits, is widely developed in the southeast of the mining area. There are three main kinds of understanding about the genesis of fluorite ore in this deposit: (1) it is mainly a Stratabound hydrothermal sedimentary genesis associated with Permian marine acidic volcanism (Li, 1985); (2) the fluorite ore underwent two stages of mineralization, volcanic deposition and hydrothermal modification (Nie et al., 2008); (3) the mineralization occurred in the Early Cretaceous and was formed by the intrusion of granitic magma (Xu, 2009). The potassium-argon isotopic age of 141.5 $\pm$ 1.2 Ma for altered sericite coeval with fluorite and 137.6 $\pm$ 1.1 Ma for altered illite suggest that the fluorite ore at the Suchar deposit is a product of Yanshanian magmatism (Xu, 2009). This study show that Pb–Zn mineralization at the Sucha deposit is significantly earlier than fluorite mineralization and likely destroyed by hydrothermal fluids that later formed fluorite, forming brecciated or lenticular ores. Combined with the geotectonic evolution of the deposit, this paper suggests that the Pb–Zn veins at the Sucha deposit may be the product of SEDEX-type mineralization associated with large-scale acidic magmatism in an Early Permian paleo-Asian intra-oceanic extensional tectonic setting.

6. Conclusions

Based on a comprehensive geological survey, and analysis of the in situ trace element composition of sphalerite and pyrite, we constrained the distributions and substitution mechanisms of trace elements, the temperature of the Pb–Zn mineralization, and the ore genesis of the Sucha deposit. The main conclusions of this study are.

(1) Sphalerite of the Sucha deposit is mainly riched in Fe, Mn, Cd, Co, Ni, Cu, In, while poor in Sb, Mn. Pyrite is mainly riched in Ni, As, Sb, Cr, Mn, Co, Cu, Sn, Pb, Tl, Se, while poor in Cd, Au, Ga, In, Bi.
(2) The elements Fe, Mn, Cd, In, Co, Ni, Cr, Bi and Cu in sphalerite and Ni, As, Sb, Mn, Co, Cu, Ni, As, Pb, Tl, and Se in pyrite are generally lattice-bound elements, while Pb, Cd can be present both in solid solution and/or as discrete micro/nano-inclusions. Sn enter as micro-inclusions into the sphalerite and pyrite. The substitution mechanisms include $2Zn^{2+} \leftrightarrow Fe^{2+} + Mn^{2+}$, $Zn^{2+} \leftrightarrow$

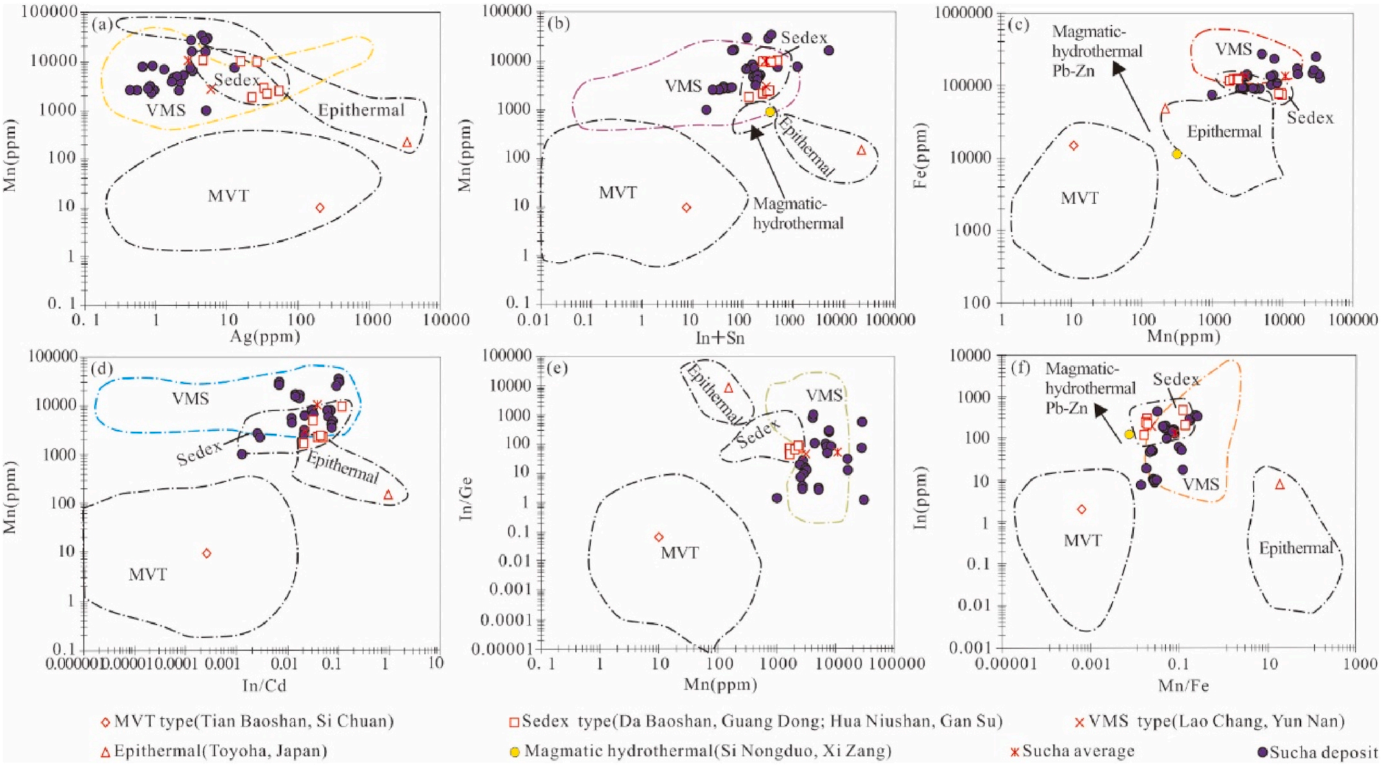

Fig. 9. LA-ICP-MS Trace element discrimination diagrams of sphalerite in the Sucha deposit (base map according to Oyebamiji et al., 2022)
Data for MVT, SEDEX, VMS, Epithermal and Magmatic hydrothermal are from Ye et al. (2017), Kang et al. (2020) and Ye et al. (2012), Ye et al. (2012), Cook et al. (2009a), and Xiao et al. (2003), respectively.

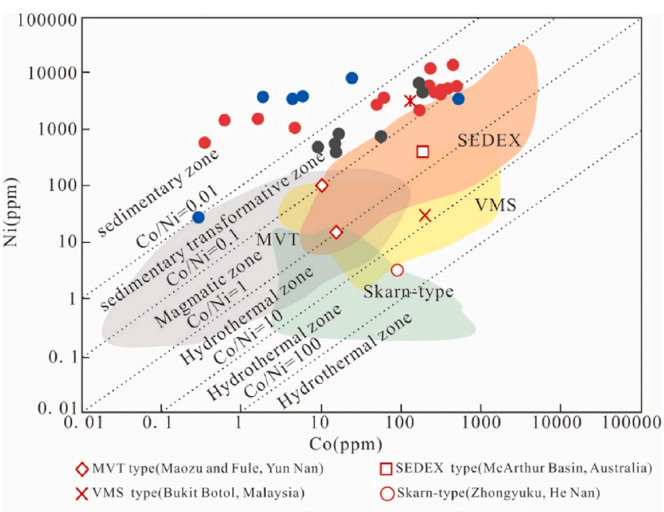

Fig. 10. Binary plots of Co vs. Ni of the Sucha deposit.
Data sources: SEDEX, VMS, Skarn and MVT from Mukherjee and Large (2017), Basori et al. (2018, 2018), Leng (2017) and Li et al. (2019b).

Co^{2+}, Ni^{2+}, $2(Ag, Cu)^{+}$+Sn^{+}↔ $3Zn^{2+}$ and $2Zn^{2+}$ ↔ Cu^{+}+ In^{3+} of sphalerite, and Fe^{2+} ↔ Co^{2+}, Ni^{2+} and (Tl^{+} + Cu^{+} + Ag^{+}) + (Sb^{3+}, As^{3+}) ↔ $2Fe^{2+}$ of pyrite.

(3) Trace element characteristic indicate that the sphalerite in this deposit formed under medium temperature conditions, while the pyrite shows a low temperature genesis. Both sphalerite and pyrite are enriched in both medium-high temperature elements and low temperature elements, suggesting that the mineralization of this deposit may have undergone a rapid cooling process from medium-high temperature to low temperature.

(4) The basic geological characteristics and trace element composition suggest that the newly discovered Pb–Zn vein can be interpreted as SEDEX type.

Authors agreement

All the authors contributed equally to the preparation of this manuscript. All authors read and approved the fnal manuscript.

Author contributions

Biao Jiang: Conceptualization, Data acquisition, Investigation, Methodology, Writing - original draft, Writing - review & editing. Denghong Wang and Yuchuan Chen: Funding acquisition, Project administration, Supervision, Resources, Writing - review. Xiulang Pu and Yu Liu: Methodology, Software, Visualization, Writing - original draft. Wenwen Ma and Wenjun Wang: Visualization, Software, editing. Chengliang Wang, Qiang Wang and Wei Chen: Project administration, Supervision, Resources, Writing – review.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.apgeochem.2023.105801.

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